

Nicholas Ashford

Software Engineer · AI Systems, Agentic Infrastructure & Distributed Systems

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I'm a software engineer and systems programmer with 18+ years of experience, currently CTO of Education Intelligence, the research business behind Teacher Tapp and School Surveys, where the platform is built in TypeScript and Rust. Over the past year I've rebuilt the company to be AI-native: agents now implement our tickets, sit in our planning calls, and connect to every system in the business. I'm always happy to be handed a challenge or something broken, and to take it apart to find out exactly why.

Highlights

- **Agentic delivery:** built a multi-agent software factory pipeline, with productivity gains that have let the business absorb natural attrition without replacement hires.
 - **Voice agent:** joins project management calls and raises contextualised tickets from the conversation.
 - **MCP infrastructure:** AI in the hands of everyone in the business, with custom MCP connectors and proxies for the systems that lacked them.
 - **Custom integrations:** energy-provider integrations at Switcher.ie and uSwitch, on-prem data extraction inside schools, MCP proxies for closed SaaS products.
 - **Systems programming:** fault-tolerant, idempotent microservices in Go and Rust that absorb extremely spiky traffic; real-time Navier-Stokes fluid simulation on the PlayStation 3 GPU.
 - **Infrastructure:** owned cloud infrastructure across AWS, Google Cloud and Azure, built with Terraform and Kubernetes.
 - **Early CI:** built an automated pull-request-to-production pipeline before mainstream CI tooling existed.
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Relevant Experience

Lead Engineer & CTO

Education Intelligence (Teacher Tapp & School Surveys)

January 2021 – Present

Education research business running the UK's largest daily teacher survey panel: thousands of teachers respond every day, and the resulting data is used by policymakers, academics and the national press.

- Designed and built a multi-agent pipeline: a workflow of agents implements tickets from specifications and acceptance criteria, raises pull requests, and ships only after automated tests and human review.
- Built a voice-driven project management agent that joins company calls, listens to the discussion and raises contextualised tickets, replacing the Kanban board.
- Rolled AI out to every member of the business and connected its systems to Claude via MCP, including a proxy service that exposes MCP connectors for SaaS products that lack them.
- Rebuilt the Teacher Tapp app from an MVP into TypeScript and React Native, establishing test-driven development and agile delivery as standard practice.
- Took the School Surveys MVP to a production-grade platform in TypeScript and Rust.

- Architected both platforms as fault-tolerant, idempotent microservices communicating over message brokers, with queue consumers written in Go.
 - Now responsible for two Rust backend platforms and nine TypeScript applications across the two businesses.
 - Built and led a team of engineers, mentoring each of them through to more senior roles.
 - Responsible for security across both businesses: protecting sensitive survey and school data, hardening infrastructure, and handling security testing and remediation.
 - Established the end-to-end product pipeline (specification, design through a retained agency, MVP build, release and analytics) so the business can take products from concept to launch.
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Senior Software Engineer

Switcher.ie

February 2019 – January 2021 · Dublin

- Part of a small team of senior engineers building the service-oriented platform behind Ireland's leading comparison site, handling lead capture from the website.
 - Designed and built the framework for distributing leads to the utility providers.
 - Owned the infrastructure: preview, staging and production environments and their monitoring, built with Terraform and Kubernetes on AWS.
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Lead Software Engineer

Assembly Edu

January 2016 – March 2019 · London

- Helped establish the company: structured as a 50/50 partnership between NEON's investment arm and an education provider, with the platform built from the ground up.
 - Designed and built the MIS Connector, a .NET ETL system running on servers inside schools, extracting sensitive management information data and syncing it to the cloud.
 - Wrote the cloud processing pipeline that turned raw school data into a queryable database, consumed by third-party education providers for analysis.
 - Owned the infrastructure (AWS) and a substantial share of the Ruby platform.
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Senior Developer

NEON Adventures

January 2014 – January 2017 · London & New York

- Consulting and tech leadership on client projects for a venture studio, spanning charities and consumer brands.
 - Built interactive 3D visualisations for a professional cycling team's website, at a time when 3D in the browser was still rare.
 - Built backends for charity clients including kindness.org and charity: water, plus a client-management system for a further charity.
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Senior Graphics Programmer

Sony Computer Entertainment Europe

February 2012 – January 2014 · London

- Graphics programmer on Harry Potter titles for PlayStation 3 (C++/GPU), on the team responsible for special effects and integrating the art-tool pipeline.
 - Built a real-time fluid simulation for the potion cauldron, implementing the Navier-Stokes equations in shaders on the PS3 GPU.
 - Implemented augmented-reality features using the PlayStation Eye, projecting a handheld device into in-game 3D environments.
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Senior Developer

Forward Internet Group (uSwitch)

July 2008 – February 2012 · London

- Full-stack development on uSwitch.com's gas and electricity comparison products: lead capture, processing and integrations with energy providers.
 - Built the company's internal deployment system: pull requests from up to 100 engineers triggered an automated pipeline through to production.
 - Founded and ran an iOS games studio within Forward's incubator: hired the team, then led development, release and marketing of a C++ iOS game over the course of a year.
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Skills & Technologies

- **AI & agentic engineering:** LLM orchestration and prompt engineering, agentic development workflows, multi-agent pipelines, MCP connector and proxy development, generative AI, machine learning, PyTorch
 - **Languages:** TypeScript, Go, Rust, C++, Python, JavaScript, C#, Ruby, Java, SQL
 - **Cloud & infrastructure:** AWS, Google Cloud Platform, Azure, Terraform, Kubernetes, event-driven architecture, message brokers, CI/CD pipelines, preview/staging/production environment provisioning, monitoring
 - **Graphics & simulation:** GPU and shader programming, real-time fluid simulation, 3D mathematics, augmented reality
 - **Data & integrations:** ETL pipelines, on-premise data extraction, third-party API integrations, Microsoft SQL Server, large-scale data processing
 - **Web & mobile:** React, Node.js, .NET, iOS, Android
 - **Leadership & practice:** team building and mentoring, product strategy, board reporting, agile delivery, test-driven development, service-oriented architecture
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Location

British citizen currently based in Dubai, with EU work rights through spouse.